

units - practice problems

name _____

1. A trail sign says the lake is 3.4 km away. How far is that in meters?

2. Convert 0.075 km to meters.

3. A lab table is listed as 180 cm long in an old inventory sheet. How long is it in meters?

4. A AA battery is about 50 mm long. How long is it in meters?

5. In a dim room, a phone camera keeps its shutter open for 12 ms to take a photo. A blink takes about 100 ms. How long is the shutter open in seconds?

6. Convert 350 μs to seconds.

7. Red light from a small laser pointer has a wavelength of about 650 nm. How long is that in meters?

8. In a straight line, New York is about 3.9 Mm from Los Angeles. How far is that in meters?

9. Convert 0.006 m to millimeters.

10. The 1500 m is a common track race. How many kilometers is it?

11. Convert 0.0045 s to milliseconds.

12. Convert 0.000 003 2 s to microseconds.

13. A bike wheel turns 250 times during a 3 minute ride. How far did the bike travel in meters?

14. A water bottle holds 0.75 L. How much is that in cubic meters? (1 L = 0.001 m^3)

15. Write 12 kg m/s^2 using the special unit abbreviation.

16. Convert 18 minutes to seconds.

17. A movie is 2.5 hours long. How many seconds is that?

18. Convert 5400 s to hours.

19. A car is moving 72 km/hr on a road with a 55 mph speed limit sign nearby. How fast is the car moving in m/s?

20. Convert 25 m/s to km/hr.

21. Convert 55 miles per hour to meters per second. (1 mile = 1.6 km)

22. In 2009, Usain Bolt set the 100 m world record with a time of 9.58 s. What was his average speed in m/s and in km/hr?

23. A highway speed limit sign in Canada says 100 km/hr. Is that faster or slower than a 65 mph speed limit in the US? (1 mile = 1.6 km)

24. A 5 km race is also called a "5K." How many miles is it? (1 mile = 1.6 km)

25. A student is 5 feet 4 inches tall. How tall are they in meters? (1 inch = 2.54 cm)

26. TV screens are measured diagonally. A TV is advertised as 55 inches. How big is the screen in centimeters? (1 inch = 2.54 cm)

27. An American football field is 100 yards long between the goal lines. How long is that in meters? (1 yard = 0.914 m)

28. A weather app shows a temperature of 18°C, wind of 15 km/hr from the west, air pressure of 101 kPa, and 2 mm of rain today. Which of these are vectors, and which are scalars?

29. A rescue boat leaves the dock and travels 3.0 km north, then 1.5 km east. How far is the boat from the dock?

30. A soccer ball is kicked at 20 m/s at 30° above the ground. How fast is it moving horizontally and vertically?

31. A wheelchair ramp is 4.0 m long and slopes up at 5° above the horizontal. How high does the ramp rise?

32. A displacement is 120 m at 60° above the x-axis. Find the x and y components.

33. A hiker walks 50 m at 30° north of west. How far west and how far north did the hiker go?

34. A vector has an x-component of 12 m and a y-component of 5 m. What are the vector's magnitude and angle above the x-axis?

35. Earth's diameter is about 1.27×10^7 m. How many megameters is that?

